SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

# ENG201

**Timed Online Assignment - July Semester 2022**

## Linear Systems Analysis and Design

**Monday, 14 November 2022** **10:00 am - 12:00 pm**

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**Time allowed: 2 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your Name and Student PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT space.
   Example, TOAENG201_RaphaelLee_T1923161 (omit D/O, S/O).

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6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed in Times New Roman font, size 12 and single spacing. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. Flowcharts and graphs may be scanned or photographed and embedded in the Word file in the right sequence provided it does not exceed the file size limit of 500MB. All uploaded handwritten answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

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Answer all questions (100 marks)

**Question 1a**

The following signals exist for all time:

- $\cos 3\pi t + \sin 7\pi t$
- $\cos 2\pi t + \sin 3t$
- $[e^{-5t}]u(t)$

**Question 1a(i)**

Analyse which of the above signals are periodic. Calculate the time period corresponding to the periodic signals.

(3 marks)

**Question 1a(ii)**

Analyse which of the above signals are power signals. Assuming the above signals to be voltage signals applied to a 1Ω resistor, calculate the average power corresponding to the power signals.

(3 marks)

**Question 1a(iii)**

Analyse which of the above signals are energy signals. Calculate the energy corresponding to the energy signal.

(4 marks)

**Question 1b**

Consider the following discrete-time signal

$$x[n] = \alpha^n u[n],$$

where $u[n]$ is a unit step function. Is this an energy signal? If so, what is its energy?

(5 marks)

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**Question 1c**

A LTI system has an impulse response as follows:

$$h(t) = e^{-i4t}u(t).$$

This system is excited by an input signal

$$x(t) = 5\sin\left(3t + \tan^{-1}\left({}^{3}/_{4}\right)\right).$$

Describe the output signal using appropriate mathematical expression.

(10 marks)

**Question 2a**

Determine whether the following systems are time-invariant or time-variant:

**Question 2a(i)**

$y(t) = \frac{1}{C}\int_{-\infty}^{t} x(\tau)d\tau$, where $C$ is capacitance of a capacitor.

(3 marks)

**Question 2a(ii)**

$y[n] = x[n]x[n+1]$.

(3 marks)

**Question 2b**

The output from a discrete-time system is given by

$$y[n] = \sum_{k=-\infty}^{\infty} 2^{(k-n)}x[k+1],$$

where $x[k]$ is the input signal.

**Question 2b(i)**

Discuss the causality of the above system.

(4 marks)

**Question 2b(ii)**

Find the impulse response of the system.

(6 marks)

**Question 2c**

It is known that the Fourier Transform of a signal $p(t)$ is given by

$$P(\omega) = \frac{j\omega}{(1 - \omega^2 + j\omega)}.$$

Solve the Fourier Transform of the following signals.

**Question 2c(i)**

$p(t-1)$

(3 marks)

**Question 2c(ii)**

$p\left(\frac{t}{2} - 2\right)$

(3 marks)

**Question 2c(iii)**

$\frac{d}{dt}(p(-2t))$

(3 marks)

**Question 3a**

It is known that the Laplace Transform $Q(s)$ of a system has a pair of simple poles at $s = 0$ and $s = -10$ and a single zero at $s = -20$. It is also known that:

$$\lim_{t \to 0} q(t) = 20.$$

**Question 3a(i)**

Find $Q(s)$.

(4 marks)

**Question 3a(ii)**

Find $q(t)$.

(3 marks)

**Question 3a(iii)**

Sketch $q(t)$ indicating the final value.

(3 marks)

**Question 3b**

A linear time-invariant system is governed by the following differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = x(t),$$

where $x(t)$ and $y(t)$ are input and output signals, respectively To this system a signal $x(t) = Ke^{-3t}$ is applied at $t = 0$ Assuming the following initial conditions:

$$y(t) = 2 \text{ at } t = 0,$$

$$\frac{dy(t)}{dt} = -1 \text{ at } t = 0.$$

Solve for the value of $K$ if the output signal $y(t) = 1.5$ for 1 second after the application of the input signal.

(15 marks)

**Question 4**

A causal linear time-invariant discrete-time system is described by the following difference equation.

$$y[n] = y[n-1] + y[n-2] + x[n-1],$$

where $x[n]$ is the input signal.

**Question 4a**

Draw a block schematic for the above given system, indicating clearly the various signals around the system.

(5 marks)

**Question 4b**

Using Z-Transform, find the system transfer function and hence determine the poles and zeros of the system.

(10 marks)

**Question 4c**

Calculate the impulse response of the system and comment on the system's stability.

(10 marks)

**----- END OF PAPER -----**